

```

import nltk

nltk.download('punkt_tab')

[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.

True

nltk.download('stopwords')

[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.

True

nltk.download('wordnet')

[nltk_data] Downloading package wordnet to /root/nltk_data...

True

# performing tokenization
#both word and sentence

from nltk.tokenize import word_tokenize,sent_tokenize

text="I AM A GOOD GIRL THOUGH. Is your habits good?"

x=word_tokenize(text.lower())
print(x)

['i', 'am', 'a', 'good', 'girl', 'though', '.', 'is', 'your',
'habits', 'good', '?']

y=sent_tokenize(text)
print(y)

['I AM A GOOD GIRL THOUGH.', 'Is your habits good?']

#REMOVING STOPWORDS AND PUNCTUATIONS(? ; : " ' {} [] ())

#FIRST PRINT WHAT ARE STOPWORDS IN ENGLISH

from nltk.corpus import stopwords
z=set(stopwords.words('english'))
print(z)

{'aren't', 'as', "it's", 'shan', 'more', 'won', 'but', 'not',
'themselves', "we've", "needn't", 'once', 'my', "she'd", 'a', "we'll",
'under', "mightn't", 'other', 'them', 'your', 'has', "couldn't",
'don', 'any', 'at', 'of', 'ain', 'is', 'yours', 'had', 'or', 'over',
'those', 've', "don't", 'having', 'down', 'where', 'theirs', 'below',
'both', 'hers', 'couldn', 'herself', "mustn't", 'if', 'each', 'such',

```

```
'to', 'an', "doesn't", "should've", 'isn', 'should', 'off', 'the',
'then', 'himself', 't', "wasn't", 'before', 'by', 'it', 'now',
"they're", "won't", 'itself', 'll', 'can', 'she', 'shouldn', 'wasn',
"he'd", 'mightn', 'y', "you've", 'being', 'for', "it'd", 'didn', 'm',
"hadn't", 'most', 'because', 'from', 'o', 'does', "i'm", 'until',
"he's", 'too', 'above', 'out', 's', 'be', 'have', 'been', "i've",
'were', 'here', 'against', 'his', 'me', 'what', 'with', 'doesn',
'its', 'few', 'whom', 'into', 'ma', "i'd", 'mustn', "they'll",
"it'll", "we'd", 'again', 'own', 'about', 'you', 'do', 'doing',
"shouldn't", 'than', 'd', 'further', "they'd", 're', 'how', "hasn't",
'ourselves', 'after', 'did', "isn't", 'him', 'there', 'up', 'ours',
'and', "shan't", "we're", 'who', 'yourselves', 'this', 'through',
"you're", 'we', 'that', 'just', 'needn', 'during', 'their', 'while',
'only', 'was', "i'll", 'he', 'wouldn', "didn't", 'yourself', "she'll",
'hadn', 'i', "you'll", 'some', 'will', 'why', 'same', 'all', 'very',
"she's", 'on', 'so', 'they', 'between', 'her', 'hasn', "wouldn't",
'are', 'haven', "he'll", "that'll", 'in', 'aren', 'nor', 'when',
"weren't", "haven't", 'our', 'these', 'weren', 'am', "they've",
"you'd", 'which', 'no', 'myself'}
```

```
import re
```

```
filtered_text=[]
for i in x:
    if i not in z:
        filtered_text.append(i)

print("TOKENIZED SENTENCE\n",x,"\n")
print("AFTER REMOVING STOPWORDS\n",filtered_text)
```

```
-----
-----
NameError                                Traceback (most recent call
last)
```

```
<ipython-input-1-0c880b9c2e20> in <cell line: 0>()
      1 filtered_text=[]
----> 2 for i in x:
      3     if i not in z:
      4         filtered_text.append(i)
      5
```

```
NameError: name 'x' is not defined
```

```
ft=[]
for i in x:
    if i not in z:
        ft.append(i)
```

```
print("TOKENIZED SENTENCE\n",x,"\n")
print("AFTER REMOVING STOPWORDS\n",ft)
```

TOKENIZED SENTENCE

```
['i', 'am', 'a', 'good', 'girl', 'though', '.', 'is', 'your',
'habits', 'good', '?']
```

AFTER REMOVING STOPWORDS

```
['good', 'girl', 'though', '.', 'habits', 'good', '?']
```

#PERFORM STEMMING -LESS ACCURATE- root word

#PERFORM LEMMATIZATION-MORE ACCURATE AND SLOWER-base word

```
from nltk.stem import PorterStemmer
ps=PorterStemmer()
words=["waits","waiting","waited","wait"]
for i in words:
    rw=ps.stem(i)
    print(rw)
```

```
wait
wait
wait
wait
```

```
from nltk.stem import WordNetLemmatizer
wl=WordNetLemmatizer()
for i in words:
    bw=wl.lemmatize(i)
    print(bw)
```

```
wait
waiting
waited
wait
```